Review of Scientific Instruments

RESEARCH ARTICLE | AUGUST 21 2014

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Rev. Sci. Instrum. 85, 11D854 (2014)
https://doi.org/10.1063/1.4893352

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Optics design for J-TEXT ECE imaging with field curvature adjustment lens[a)]

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(Presented 3 June 2014; received 31 May 2014; accepted 6 August 2014; published online 21 August 2014)

Significant progress has been made in the imaging and visualization of magnetohydrodynamic and microturbulence phenomena in magnetic fusion plasmas. Of particular importance has been microwave electron cyclotron emission imaging (ECEI) for imaging T_e fluctuations. Key to the success of ECEI is a large Gaussian optics system constituting a major portion of the focusing of the microwave radiation from the plasma to the detector array. Both the spatial resolution and observation range are dependent upon the imaging optics system performance. In particular, it is critical that the field curvature on the image plane is reduced to decrease crosstalk between vertical channels. The receiver optics systems for two ECEI on the J-TEXT device have been designed to ameliorate these problems and provide good performance with additional field curvature adjustment lenses with a meniscus shape to correct the aberrations from several spherical surfaces. © *2014 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4893352]

I. INTRODUCTION

Since the development of microwave imaging technology, multi-dimensional visualization techniques have been widely used in fusion plasma diagnostics. Electron cyclotron emission imaging (ECEI) diagnostic systems have thus far been installed on numerous tokamaks including TEXTOR,[1] KSTAR,[2] DIII-D,[3] ASDEX-U,[4] EAST,[5] HL-2A, and HT-7.[6]

Field curvature concerns the shape of the focal plane created by the effect of the imaging optics upon an imaging array. The effects of field curvature upon microwave imaging reflectometry (MIR)[7] performance has long been recognized, where the focal plane should be curved to match the shape of the plasma cutoff layer so that the incident waves are normally incident to the layer. The field curvature of a particular system can be examined by setting up the optics and performing Gaussian beam propagation simulations. In the case of ECEI, a flat image plane is required for optimum ECEI operation in order to robustly image radiation emitted from the flat electron cyclotron resonance layer. Field curvature results in a degradation of the resolution across the focal plane. Thus, the image of the plasma fluctuations will be destroyed by an optics system with uncontrolled field curvature, i.e., a focal plane unmatched to the inherent shape of the surface being imaged.

II. FIELD CURVATURE CORRECTION IN ECEI

The receiver optics for ECEI collects electron cyclotron emission (ECE) from the plasma, focusing the ECE radiation onto a linear detector array while achieving high spatial resolution along the poloidal axis. The emission surface for a particular emission frequency is solely a function of the magnetic field profile of the plasma being imaged. For most tokamak plasmas this surface is effectively flat on a line of constant major radius. When the focal plane has finite field curvature and is not aligned to the object plane (emission layer in plasma side), the effective spot sizes across the object plane can vary significantly, which is clearly shown in Figure 1(a). The field curvature of the focal plane results in a radial distance between the waists of the center and edge sightlines. For the simple case shown in Figure 1(a), the sample volume sizes of the edge channels will increase, which causes the deterioration of the spatial resolution and possible crosstalk between adjacent vertical channels.

The importance of the field curvature issue in the optical design of ECEI system did not attract enough attention in the old days, since the ECEI systems were usually applied to investigate the large spatial magnetohydrodynamic (MHD) mode structure. An instance is the optics of the ECEI system on HT-7 tokamak, which comprises a big substrate lens in front of the antenna array, resulting in a strong field curvature, about 80 mm (see Fig. 1(b)). For the observation of small-scale spatial structures, the strong field curvature will lead to serious distortion on ECEI image. In order to reuse most optical components in the old HT-7 optics for the J-TEXT[8] ECE Imaging experiment, it is necessary to reduce the field curvature to improve image quality. New imaging optics need to

[a)]Contributed paper, published as part of the Proceedings of the 20th Topical Conference on High-Temperature Plasma Diagnostics, Atlanta, Georgia, USA, June 2014.

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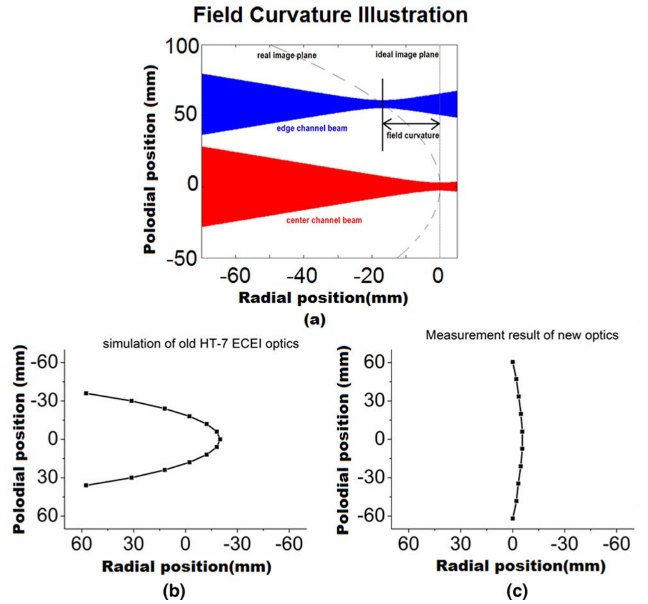

FIG. 1. (a) An example with the field curvature, in which the radial distance between waists of the center sightline and edge sightline is defined as field curvature on the image plane. (b) Simulation result of HT-7 ECEI image plane with a field curvature about 80 mm. (c) The measurement result of the optimized optics with two additional FCA lenses.

be developed to correct this aberration to achieve an ideal flat image plane. Two additional meniscus field curvature adjustment (FCA) lenses are used for the new optics on J-TEXT ECEI with the old HT-7 detector array. According to the simulation results shown in Figure 1(c), the strong field curvature is greatly reduced to about 10% of the Rayleigh length with these two FCA lenses.

A. Optics system for ECEI on J-TEXT

The total J-TEXT ECEI system is composed of 384 channels with two antenna arrays toroidally separated by 67.5° (see Fig. 2). For each array box, there are 12 poloidal channels with 16 channels in the radial direction. Both of the optical systems are oriented for perpendicular observation. It should be noted that the spaces between adjacent antennas in the two antenna arrays are different. For the old HT-7 ECEI array[9] applied in the port 2 system, which has one whole large hyper-hemispherical substrate lens in front of the antennas, the space between adjacent antennas is small, corresponding to a congenitally fine poloidal resolution. The second antenna array in the port 5 system utilizes separate mini-lenses[10] attached in front of each antenna, caused a larger space between adjacent antennas.

We compare the two optics system (the old optics for HT-7 and the revised optics for J-TEXT) with same detector array and same hyper-hemispherical substrate lens. Besides the substrate antenna lens, the old optics used on HT-7 had only three lenses, shown in Fig. 3(a). In the revised ECEI optics for J-TEXT, two FCA lenses are included. The first lens is set next to the substrate lens which is fixed in the antenna array box. The second one is placed between the zoom lenses (see Fig. 3(b)). The field curvature can be finely-tuned by moving the second FCA lens. In this optimized configuration, the

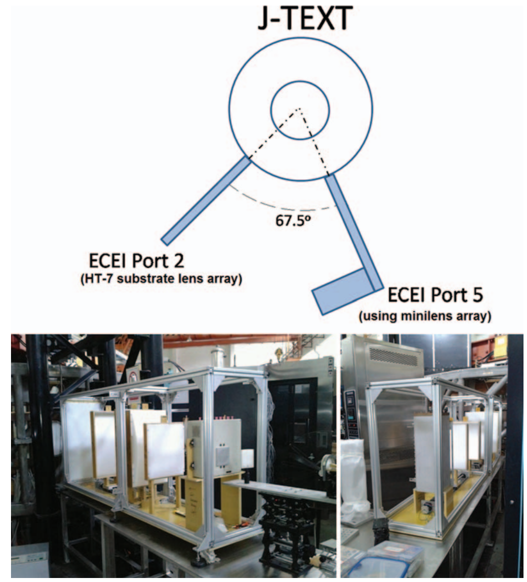

FIG. 2. Top view of two ECEI systems on toroidally separated windows on J-TEXT. These two ECEI receiver optics systems with substrate lens array from HT-7 ECEI (left) and mini-lens array (right), respectively, are installed on J-TEXT.

image shape can remain flat while focusing on different radial positions in the plasma. Shown in Fig. 4, the focusing position of the imaging plane can be adjusted in a wide radial range with a constant flat field curvature.

B. Performance parameters

The two front-end optics systems on J-TEXT were designed to match the flat emission layers over the entire radial extent of the plasma. Both of them have adopted FCA lenses for distortion correction on image plane.

The spacing between two adjacent antennas on the substrate lens is 2.032 mm, which is the same as the waist in the Gaussian beam calculation. As clearly shown in Fig. 3(b), there are one H-lens and five E-lenses in the optical path. Two

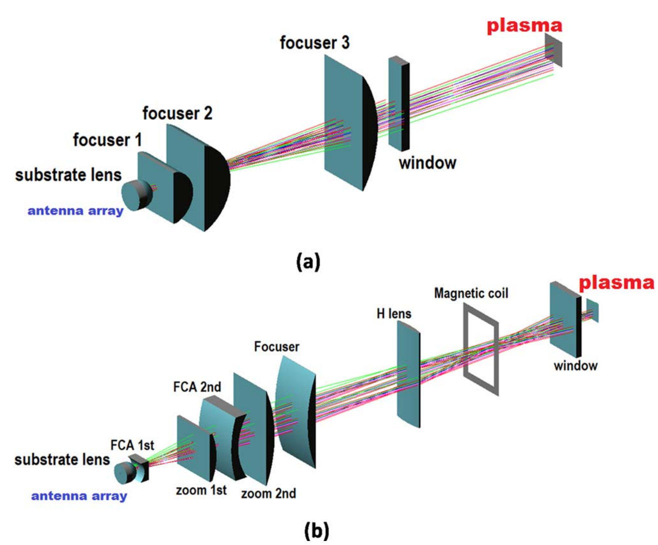

FIG. 3. Three-dimensional optics renderings of the old optics on HT-7 (a) and the optimized optics with additional FCA lenses for J-TEXT (b).

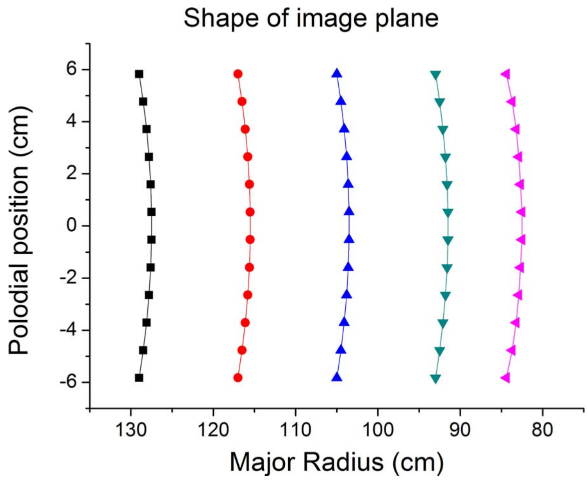

FIG. 4. Shapes of the image plane of the J-TEXT substrate lens optics system at different focusing positions.

of E-lenses, FCA 1st and 2nd, are meniscus lenses used to adjust the shape of the focal plane. The position of the focal plane covers the range in normalized radius from −0.98 to 0.98 by moving the "focuser lens." Consider the optical thickness and typical magnetic field on J-TEXT, the reasonable radial range for X2 ECE is from −0.61 to 0.74. The zoom factor range can be changed from 1.3 to 3.5 using several zoom options for large scale or high-resolution observations. The shape of the focal plane remains flat for different focusing positions.

Same as the port 2 system, the optical system on port 5 with fine spatial resolution has good performance to cover the q = 1 flux surface. The optical system was designed to cover the same imaging area in the plasma as covered by the port 2 system. In order to match the size of the separate mini-lenses employed in port 5 system, the waist in the antenna side is set up to 11 mm, which will benefit to collect the emission more efficiently. With the larger space between adjacent antennas, the total length of the antenna array in port 5 system is larger than the one in port 2 system. Thus a smaller zoom factor is required for the antenna array with mini-lenses, in order to observe the plasma phenomena with small scales. The typical waist in plasma side is about 7 mm, which ensures a good spatial resolution in vertical direction.

In order to investigate the 3D characteristics of the MHD structure near the q = 1 surface, these two separate ECEI systems in different toroidally positions are focusing on the center of plasma. Characteristic parameters of the two optical systems are included in Table I. Each of them has a wide zoom range for independent operation by itself, suitable for the observations of both small and big scale structures. It should be stressed that great aberration correction has been realized in both optics with the help of additional FCA lenses.

TABLE I. Basic parameters of two ECEI optics systems with different antenna array.

	Substrate@port2	Minilens@port5
Total number of channels	192	192
Radial coverage (norm. radius)	−0.61–0.74	−0.61–0.74
Radial resolution (mm)	10 (typical)	10 (typical)
Poloidal resolution (mm)	7.5 (typical)	7.2 (typical)
Poloidal zoom factor	1.57–3.70	0.56–1.1
	3.70 (typical)	0.65 (typical)
Poloidal coverage (mm)	100–210	150–325
Field curvature (mm)	**4.5–5.6**	**0.48–0.51**
Rayleigh length (mm)	56.3 (typical)	51.8 (typical)

III. SUMMARY

Field curvature results in aberrations in large aperture microwave imaging optics systems which reduce the ECEI spatial resolution in 2D electron temperature fluctuations measurement. The adoption of FCA lenses can correct the shape of the image plane, resulting in flat surfaces to match emission layers in plasma. Two optics systems with additional FCA lenses using different types of detector arrays have been designed and fabricated for J-TEXT experiment, in which the field curvature has been greatly reduced to about 10% of the Rayleigh length and ideal flat imaging plane is almost realized.

ACKNOWLEDGMENTS

This work was supported by National Magnetic Confinement Fusion Energy Program of China under Contract Nos. 2009GB107001 and 2014GB109002, and by the U.S. DOE Grant No. DE-FG02-99ER54531.

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